

Exercise 02

1. $x6 = 2$
2. If register $x6$ is initialized to 10, and $x5$ to zero.
 - a. The final value of $x5$ is 20.
 - b. C code:

```
1      acc = 0;
2      i = 10;
3      while (i != 0) {
4          acc += 2;
5          i--;
6      }
7
```

- c. $4*N + 1$ instructions.
- d. (Note: change condition $!=$ to $>=$ in the while loop)

```
1      acc = 0;
2      i = 10;
3      while (i >= 0) {
4          acc += 2;
5          i--;
6      }
7
```

3. This C code corresponds most directly to the given assembly.

```
1      int i;
2      do{
3          result += MemArray[i * 2];
4          i++;
5      }while (i < 100;)
```